

**B.Tech. Degree II Semester Regular and I Semester Supplementary
Examination May 2016**

**IT/CS/EC/CE/EE/ME/SE 1105 B /1205 A BASIC ELECTRONICS ENGINEERING
(2015 Scheme)**

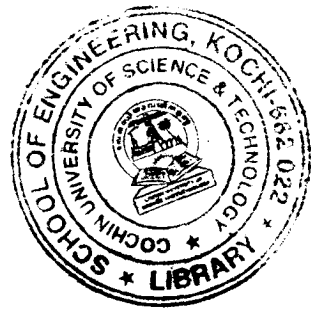
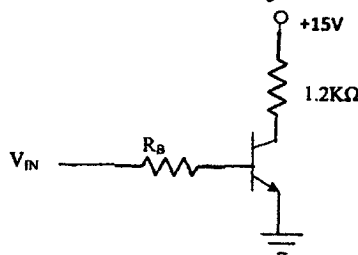
Time : 3 Hours

Maximum Marks : 60

**PART A
(Answer ALL questions)**

(10 × 2 = 20)

- I. (a) Differentiate drift and diffusion currents.
 (b) A full wave rectifier produces an 80V peak rectified voltage from a 60Hz AC source. If $10\mu F$ capacitor filter is used, determine the ripple factor for a load resistance of $10K\Omega$.
 (c) The transistor in the figure shown below has a β_{DC} of 50. Determine the value of R_B required to ensure saturation when V_{IN} is 5V. Assume $V_{CEsat} = 0V$ and voltage across the forward biased base emitter junction as 0.7V

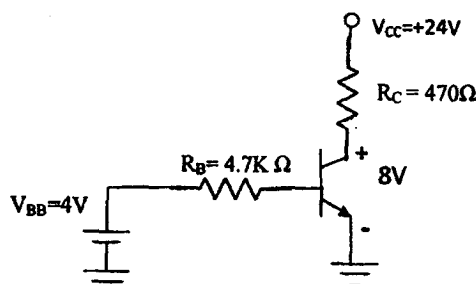


- (d) Explain the Barkhausen criteria for oscillation.
 (e) What is the primary advantage of switching regulators over linear regulators?
 (f) Convert octal 256 to hexadecimal number.
 (g) What is a microprocessor? Give its functions.
 (h) Explain sampling theorem.
 (i) What is the bandwidth of an amplitude modulated wave if the carrier signal of frequency f_c is amplitude modulated by a sinusoidal signal of frequency f_m ?
 (j) What are the principle, merits and limitations of FM.?

PART B

(4 × 10 = 40)

- II. (a) Explain the working of a zener diode.
 (b) Determine each current in figure shown below. What is β_{DC} .



OR

(P.T.O.)

- III. (a) Explain the working of full wave centre tapped rectifier, with neat sketches. Explain the effect of capacitor filter on the output of the circuit.
 (b) The rms output voltage of a bridge rectifier is 20V. What is the peak inverse voltage across the diodes?

- IV. (a) Draw the circuit of common emitter amplifier and explain the principle of amplification. Also explain the need of each component in the circuit.
 (b) Compare the characteristics of CB, CE and CC amplifiers.

OR

- V. (a) Explain the working of transistor LC oscillator.
 (b) In a transistor RC phase shift oscillator, the resistors and capacitors used in the circuit are equal and they are $10\text{ K}\Omega$ and $0.001\text{ }\mu\text{F}$ respectively. Calculate the frequency of oscillation.

- VI (a) What is the difference between combinational circuits and sequential circuits?
 (b) Construct two input NAND gate using NOR gates only.
 (c) Using Boolean law, prove the following.

$$AB + \bar{A}C + BC = AB + \bar{A}C$$

OR

- VII. (a) Explain the working of CRO, with the help of block schematic.
 (b) Describe the working of a sensor for force measurement.

- VIII. (a) How are the signals classified? Explain.
 (b) Describe the properties of transforms.

OR

- IX. (a) Derive an expression for amplitude modulated wave. What is the modulation index?
 (b) What do you understand by angle modulation? What are the types of angle modulation?

